Solar PV Systems Questions and Answers

SECTION A (40 MARKS)

1. Advantages of Solar Energy Over Other Forms of Energy

- Renewable and sustainable
- Reduces electricity bills
- Environmentally friendly (zero emissions)

2. Considerations for Selecting a Solar Module

- Efficiency
- Durability and warranty
- Cost

3. Main Classifications of Inverters

- Pure sine wave inverters
- Modified sine wave inverters

4. Considerations for Selecting Batteries

- Capacity and power rating
- Depth of discharge (DoD)
- Battery lifespan
- Maintenance requirements

5. Definition of Depth of Discharge

- The percentage of a battery's capacity that has been used.

6. Main Types of Solar PV Modules

- Monocrystalline silicon
- Polycrystalline silicon
- Thin-film solar cells

7. Types of Solar Radiation

- Direct solar radiation
- Diffuse solar radiation

8. Routine Maintenance Practices for Solar PV Systems
- Cleaning solar panels
- Inspecting wiring and connections
- Checking battery health
- Ensuring inverters function properly
- Monitoring system performance

9. Explanation of Solar Cells
- A solar PV module consists of many small solar cells that convert sunlight into electricity.

10. Technologies for Harvesting Solar Energy
- Photovoltaic (PV) cells
- Solar thermal systems
- Concentrated solar power (CSP)

SECTION B (60 MARKS)
11(a). Types of Charge Controllers
- PWM (Pulse Width Modulation)
- MPPT (Maximum Power Point Tracking)
- Simple ON/OFF controllers

11(b). Factors to Consider When Selecting Solar Components
- Charge Controller: Type, voltage rating, efficiency
- Battery: Type, capacity, maintenance requirements
- Inverter: Type (pure/modified sine wave), capacity, efficiency

Question 12
(a) Economic Considerations for Installing a Solar PV System
- Initial investment cost
- Return on investment (ROI)

- Government incentives
- Maintenance costs

(b) Factors Affecting Solar PV Output
- Sunlight intensity
- Temperature effects
- Shading
- Panel orientation and tilt angle

(c) Characteristics of a Good Solar Battery
- High storage capacity
- Long lifespan
- High efficiency
- Fast charging and slow discharge rate

Question 13
(a) Layout of an Off-Grid Solar PV System
- Solar panels generate DC power
- Charge controller regulates power to the battery
- Battery stores energy
- Inverter converts DC to AC for use

(b) Advantages of Grid-Connected Solar PV Systems
- Lower installation cost
- Reliable power supply
- Net metering benefits
- Reduced maintenance
- Efficient energy use

Question 14
(a) Calculation of Solar Installation Output
- Given:
  - Power per panel = 300W

  - Voltage = 12V
  - Current = 8A
  - Efficiency = 20%
  - Number of panels = 4

- **For Series Connection:**
  - Voltage = 12V × 4 = 48V
  - Current = 8A (remains the same)
  - Total Power = 48V × 8A = 384W

- **For Parallel Connection:**
  - Voltage = 12V (remains the same)
  - Current = 8A × 4 = 32A
  - Total Power = 12V × 32A = 384W

(b) Safety Practices for Handling Solar Batteries

- Ensure proper ventilation
- Wear protective gear
- Avoid short circuits
- Regular maintenance and inspections